

Standard Form & Indices

Test 4 · 41 marks total · Sections F & G include Higher-tier content

Name: _____

Date: _____

Score: _____ / 41

Section A — Laws of indices

A1. Write $5^6 \times 5^3$ as a single power of 5.

[1 mark]

A2. Write $9^{10} \div 9^4$ as a single power of 9.

[1 mark]

A3. Write $(7^2)^4$ as a single power of 7.

[1 mark]

A4. Work out the value of $4^2 \times 4^1$.

[2 marks]

Section B — Zero and negative indices

B1. Work out the value of 15^0 .

[1 mark]

B2. Work out the value of 2^{-3} . Give your answer as a fraction.

[2 marks]

B3. Work out the value of 10^{-6} . Give your answer as a decimal.

[2 marks]

Section C — Writing standard form

C1. Write 91,000 in standard form.

[2 marks]

C2. Write 5,600,000 in standard form.

[2 marks]

C3. Write 0.0039 in standard form.

[2 marks]

C4. Write 0.00008 in standard form.

[2 marks]

Section D — Converting back, and calculating

D1. Write 3.4×10^4 as an ordinary number.

[1 mark]

D2. Write 9×10^{-5} as an ordinary number.

[1 mark]

D3. Work out $(6 \times 10^3) \times (5 \times 10^4)$. Give your answer in standard form.

[3 marks]

D4. Work out $(8 \times 10^9) \div (4 \times 10^4)$. Give your answer in standard form.

[3 marks]

Section E — Powers to know, and estimating roots

E1. Write down the value of 5^3 .

[1 mark]

E2. $\sqrt{130}$ lies between which two consecutive whole numbers?

[1 mark]

Section F — Laws of indices with algebra

F1. Simplify $u^8 \times u^5$.

[1 mark]

F2. Simplify $v^{12} \div v^6$.

[1 mark]

F3. Simplify $(w^5)^2$.

[1 mark]

F4. Simplify $6f^3 \times 2f^5$.

[2 marks]

F5. Simplify $(3g^2)^3$.

[2 marks]

Section G — Fractional indices (Higher)

G1. Work out the value of $64^{1/2}$.

[1 mark]

G2. Work out the value of $125^{1/3}$.

[1 mark]

G3. Work out the value of $27^{2/3}$.

[2 marks]

G4. Work out the value of $25^{-1/2}$.

[2 marks]